

## Autopsy of Five Companies

*What the bankruptcies of the blood-substitute industry say about molecules, money and the odd survival of a cow protein*

Medicine likes to tell the story of its failed products as a story about the products. The molecule was toxic, the trial was underpowered, the mechanism was misunderstood. Those explanations are sometimes correct. But they leave out a second story running in parallel, which is about the organizations that made the molecules. Companies have anatomies. They have capital structures, burn rates, single points of failure and moments at which a bad quarter turns into an existential one.

This essay reads the history of the oxygen-carrier industry as a series of corporate autopsies. I will not pretend to a level of detail that public sources do not support, and where I am uncertain I will say so. The aim is not gossip. The aim is to identify the patterns that repeat, because those patterns tell a patient community what has to be different this time if a BHOC is going to be available in five years and not merely promised.

### The peculiar economics of a product with no customers yet

Begin with the structure of the problem. A company that wants to sell an oxygen carrier for human use has to do four expensive things in sequence, and it has to pay for each before it earns a dollar.

First it has to build a manufacturing plant capable of producing a sterile injectable from an animal or recombinant source, at a scale that regulators will accept as representative of commercial supply. This is a capital project measured in tens of millions of dollars. Second, it has to run clinical trials in emergency and surgical settings, which are among the most difficult and costly trials in medicine, with sites that must screen patients in minutes and enroll them without consent. Third, it has to survive regulatory review, which for a first-of-class product with a troubled predecessor history is slow and unforgiving. Fourth, it has to build a market in hospitals that already have a working alternative in the blood bank.

The difficulty is that none of these steps is optional and none can be shortened by cleverness. The fixed costs are borne before there is a revenue stream to service them. In the financial vocabulary, the business model demands long duration equity and offers a binary outcome. Investors call this the valley of death, and it swallowed more than one company in this field.

### Five stories, told briefly

The following sketches are drawn from public reporting and reviews. They are intentionally brief and I have kept to facts I could source.

**Baxter and the diaspirin hemoglobin.** The best-known early product from a major pharmaceutical firm was a human hemoglobin cross-linked between its alpha chains, sold under the developmental name HemAssist. One published review states that in a trial in trauma patients the product produced a large increase in mortality compared with saline, and the

review's figure was a 72 percent increase. The program ended. What is notable is that this was a large, well-financed company with regulatory expertise, and the failure was scientific rather than financial. It set the tone for the entire class in the eyes of regulators and journal editors, and the memory of that trial is still invoked when new products are discussed, even though the molecule is chemically different from a bovine polymer.

**Hemosol and the raffinose-linked hemoglobin.** A Canadian company developed a human hemoglobin cross-linked with a sugar-derived reagent. According to the same review, the program advanced to late-stage trials and was discontinued in 2003 after patients receiving the product experienced adverse cardiac events. The company's own finances deteriorated in the years that followed, and it entered insolvency proceedings in the middle of the decade. Here, financial death followed scientific death by a short interval, which is the typical pattern for a company whose entire value is a single product.

**Northfield and the polymerized human hemoglobin.** This Illinois company pursued the field's most ambitious trial, the prehospital trauma study discussed elsewhere on this site, which used an exception from informed consent. The regulatory outcome was unfavorable and the pooled safety analysis published in 2008 hurt further. The company filed for bankruptcy in 2009. What distinguishes Northfield's case is that it was bold. It attempted the hardest indication first, in a setting where the comparator was standard emergency care and the endpoint was mortality. If it had succeeded, it would have transformed emergency medicine. It did not, and there was no second product to fall back on.

**Biopure and the bovine glutamers.** The Massachusetts company that developed the two bovine products, one for dogs and one for people, is the most relevant case here, and its history is documented in some detail. It was founded in 1984 and began by sourcing bovine blood from regional slaughterhouses. In 1999 the Food and Drug Administration and the European Commission approved the veterinary product for anemic dogs. In April 2001 South Africa approved the human product for surgical anemia. It was also approved in Russia. But the American regulator placed a hold on a proposed trauma study in April 2003 and rejected an application for use in orthopedic surgery in July of that year. Later that year the stock fell sharply after the Securities and Exchange Commission found misleading statements about the regulator's stance, and in 2005 the commission brought charges against the company's chief executive and other officers. In 2008 the pooled safety analysis appeared. On July 16, 2009 the company filed for Chapter 11 bankruptcy protection, and its assets were bought by a private entity that September.

**The successor entities.** The assets passed through more than one owner. By 2014 the human product was being made available through the regulator's expanded access mechanism, and in 2022 a successor company filed a registration statement seeking about 30 million dollars in an initial public offering, drawing a skeptical response from at least one financial journalist who called the attempt a long shot. I do not know how that filing ended, and I would rather say so than guess.

## The patterns

Read the five stories side by side and five patterns emerge.

**The single-product company.** Every one of these organizations was built around one molecule. When the molecule stumbled, there was nothing to absorb the shock. Big pharmaceutical companies that took on hemoglobin projects, like Baxter, could shelve the effort and continue. A small company cannot. The lesson is not that small companies should not try, but that a small company needs a structure that separates its survival from any single trial result. Options include a platform with several indications, a diversified funding base, or a partnership in which a larger organization shares the downside.

**Judgment by class.** When one product harmed patients, regulators, reviewers and journalists learned to treat the whole class with suspicion. The pooled analysis in 2008 was an example: it combined trials of several different molecules and produced a single alarming number. That is a fair scientific act, because the products shared a mechanism of concern. It was also a commercial catastrophe for the companies whose products differed. A bovine polymer is not chemically the same as a cross-linked human dimer. But a company cannot argue its way out of a class reputation in a single meeting. It has to generate its own data, patiently, over years, and it has to be able to afford the years.

**Improving comparators.** The blood supply became safer over the same period, in large part because of the diseases that first motivated the search for substitutes. Screening for viruses improved, leukoreduction became routine, and transfusion practice grew more restrained. Every improvement raised the bar for a product whose main pitch was safety from infectious disease. A product designed in the 1990s to offer protection against contamination that then became rare had a shrinking advantage.

**Governance failures.** The Biopure case includes an element that no chemistry explains. The commission's findings about misleading statements to investors and the criminal charges against officers destroyed trust in the company's public communications. It is difficult to overstate how corrosive this is. In a field where the science is already contested, the credibility of the sponsor becomes a scientific variable, because reviewers must decide how far to trust the sponsor's reading of its own data. A company that has been caught shading its statements pays a tax on every subsequent claim.

**Survival through niches.** And here is the most useful pattern of all. The bovine product did not vanish. It survived, in a battered form, because three small niches kept it alive: a veterinary market that valued it, a handful of countries that had approved it, and a compassionate access route for patients who could not receive blood. None of these markets could support a company on its own, but together they preserved the manufacturing know-how, the regulatory dossier and a trickle of clinical experience. When a new owner acquired the assets, it acquired something that a start-up could not have built: a product that had already been made, shipped and used. The lesson for planners is that niches are not a consolation prize. They are scaffolding. A niche that pays even modestly keeps a validated process running, keeps trained people employed and keeps a pharmacovigilance record accumulating, and those three things are the hardest to rebuild once lost. An ephemeral company can leave behind an enduring capability, provided someone is diligent enough to preserve it through the wreckage.

## What a durable model might look like

Suppose a patient organization were asked to design the funding and governance structure for a BHOC program that could survive the next fifteen years. What would it insist on?

I would begin with the customer. The natural early customers for an oxygen carrier are not consumers or even hospitals, but institutions that carry responsibility for people in places where blood cannot follow. Military medical commands, remote-region health services, disaster response agencies and organizations serving people who refuse blood on grounds of conscience all fit. Such buyers can fund development on contract, purchase in bulk and value a shelf-stable product for reasons the blood bank market does not. Because they pay before the commercial market matures, they reduce the valley.

Next comes the regulatory strategy. A rare-disease or orphan designation, where it applies, offers incentives that improve the economics of a small market. Expanded access, used carefully, generates a record of real-world use and can be a modest revenue source, though a regulator will not allow it to substitute for trials. Foreign approvals, which have some precedent for this product, can open smaller markets and produce clinical data, provided their standards of evidence are transparent.

Third is manufacturing. A plant that can only make one product at one scale is a fragile asset. Modular, smaller-scale facilities that can produce clinical lots and low-volume commercial supply reduce the capital at risk. I do not know which developers are pursuing this route, and I would ask them.

Fourth is data discipline. The lasting damage to the field came from unfulfilled promises and disputed interpretations. A program that pre-registers its trials, publishes negative results and invites independent analysis builds a kind of credit that money cannot buy. This is not idealism. It is a strategy for lowering the tax on every future claim.

Fifth is governance. A company whose communications have been audited by an independent committee, whose disclosures are conservative and whose leadership does not overstate its regulatory position offers an investor something rare in this sector: predictability. A patient organization has leverage here. It can decline to lend its name to promotional claims that go beyond the data, and it can say so publicly.

## The arithmetic of the valley

A worked example makes the structure vivid. The figures below are invented for illustration and do not describe any real company, but the shape is faithful to what the histories show.

Imagine a developer that needs 40 million dollars for a pilot plant, 25 million for two pivotal trials, and 8 million a year in overhead for five years before a decision. The total before any revenue is roughly 105 million dollars. Suppose the chance that the first trial succeeds is one in three, that regulators approve given success is one in two, and that the first year of sales is small. The expected value of the enterprise at the start is dominated by a one-in-six event. An investor who understands that will demand a return of many times the money, and a founder who understands it will be tempted to shade the probabilities.

Now add the peculiar feature of this field. A failure of one trial does not merely lose that trial's cost. It changes the prior for the whole class, so the probability of success for the next company falls too. In economic language, each failure imposes a negative externality on competitors. That is why capital in this sector has come in bursts and then vanished for years: the informational damage of a failure lingers long after the balance sheet is cleaned up.

The remedy is not obvious, but one piece is: reduce the amount that has to be raised at the riskiest point. A program that can reach an early human data point cheaply, through a small compassionate-use series or an ex vivo organ perfusion study where the drug never enters the patient's bloodstream, spends a few million dollars to learn something that would otherwise cost tens of millions. Cheap early information is the most valuable asset a small developer can buy.

### Leading indicators an outsider could have watched

It is easy to write an autopsy after the death. The harder and more useful task is to ask what an outside observer, such as a patient organization, could have noticed while the patient was still alive.

Watch the ratio of announcements to publications. A developer that issues frequent press statements and rarely publishes peer-reviewed data is telling you something about where its evidence lives.

Watch the language used about the regulator. Confident statements about what an agency has said or will say, unsupported by documents, are a warning. The Biopure episode with the securities regulator turned on exactly that kind of statement.

Watch the trial design. A first trial that tries to prove survival benefit in the most severe emergency setting, without earlier smaller studies, is an all-or-nothing wager. It may be brave. It is also how a company puts its whole future on a single roll.

Watch the manufacturing. If the company cannot describe its process, its lot release criteria and its supply redundancy in plain terms, its plant is either immature or its story is.

Watch the cash runway. A company with fewer than eighteen months of funding and a trial that will read out in twenty-four is a company that will negotiate its next financing from a position of weakness.

None of these indicators is proof of failure. Together they are a checklist that an advocacy group with access to public filings can apply, and it is a checklist that would have raised flags in several of the histories above.

### Where advocacy fits in the ledger

There is a temptation for advocates to see the industry's failures as an argument for more urgency and less caution. I think the history says the opposite. Every failure in this field included a moment of enthusiasm that outran the evidence: the early trial that was expanded before the signal was understood, the announcement that reached investors before regulators, the confidence that a favorable meeting meant approval.

What advocacy can offer is something the market underprices: patient legitimacy. A trial that patients trust enrolls faster and drops out less. A program with a credible community voice is more likely to obtain the community consultation that emergency research demands. An organization that has read the history and can say what went wrong last time is better placed to make sure it does not happen again.

It can also do something quieter. It can keep the record. The reason I can write this essay at all is that other people, journalists, regulators, reviewers, took the trouble to document how the companies rose and fell. Without that record, each new generation of developers would rediscover the same mistakes at the same cost. A patient organization that maintains a plain-language archive of what happened, product by product, is doing work the industry has no incentive to do.

### The uncomfortable conclusion

Five companies, three bankruptcies or insolvencies among them by my count, and one cow protein that limped on. If you had to summarize the pattern in a sentence, you could say that this field has been unusually hard on its organizations and, oddly, kinder to its molecules. The bovine glutamer outlived the company that made it.

That should reassure and warn in equal measure. It reassures because it suggests the molecule is not the weak point; a product that keeps finding a use after its sponsor collapses is doing something worth doing. It warns because a molecule without a durable steward has no future, however good it is. The next chapter will be decided less by chemistry than by whether the people who care about it, including the patients, insist on an organization worthy of the molecule.

### Further reading

Company histories are drawn from public sources including reviews of blood substitutes, the published account of Biopure and its assets, and news reporting on the 2022 registration filing by its successor. The 2008 pooled safety analysis appeared in the Journal of the American Medical Association. Specific figures, such as the 72 percent mortality increase cited for the diaspirin-linked product, come from a secondary review and should be checked against the original trial reports.